

Early Detection of Sepsis in Neonates using Thermal Imaging

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Index

[0 Index](#)

[1 Introduction](#)

[1.1 Context of the Challenge](#)

[1.2 General Problem to Be Solved](#)

[1.3 Summary of the Proposed Solution](#)

[2 Proposal of solution](#)

[2.1 Objectives](#)

[2.2 AI Methodology](#)

[2.3 Project planning and methodology](#)

[2.4 Ethical and Regulatory Issues](#)

[2.4.1 Data Privacy & Consent](#)

[2.4.2 Regulatory Classification](#)

[2.4.3 Ethical Risks and Mitigations](#)

[3 Implementation of the solution](#)

[3.1 System Overview](#)

[3.2 Device Architecture and Technical Specifications](#)

[3.2.1 Core System Components](#)

[3.2.2 Technical Implementation Framework](#)

[3.3 Technical Methods and Implementation](#)

[Stage 1: Modality Alignment](#)

[Stage 2: Pose Detection](#)

[Stage 3: Image Quality Assessment and Rejection Mechanisms](#)

[Stage 4: Temperature Analysis](#)

[Stage 5: Temporal Pattern Recognition & Alert Generation System](#)

[Validation Framework](#)

[3.4 Regulatory Compliance and Safety Implementation](#)

[3.4.1 Safety Mechanisms](#)

[3.5 Technical Limitations and Future Enhancements](#)

[3.5.1 Current Constraints](#)

[3.5.2 Proposed Technical Improvements](#)

[4 Results](#)

[4.1 Methodology](#)

[4.2 Results of the Experiments](#)

[4.2.1 Case 36: Posture Detection Failure](#)

[4.2.2 Case 47: Contextual False Alarm](#)

[4.2.3 Case 48: Optimal Detection](#)

[4.3 Qualitative & Quantitative analysis of the results](#)

[4.3.1 Qualitative Findings](#)

[4.3.2 Key Findings](#)

[5 Conclusions and open challenges](#)

[5.1 Key Achievements](#)

[5.2 Open Challenges](#)

[5.2.1 Dataset](#)

[5.2.2 Modality Alignment](#)

[5.2.3 Pose Detection](#)

[5.2.4 Temperature Analysis](#)

[5.2.5 Temporal Pattern Recognition & Alert Generation](#)

[6 References](#)

[Core Clinical Research](#)

[Thermal Imaging in Medical Applications](#)

[AI in Medical Diagnosis](#)

[AI in Pediatric and Neonatal Care](#)

1 Introduction

1.1 Context of the Challenge

Neonatal sepsis is a **life-threatening condition** and a leading cause of mortality in newborns, particularly in Neonatal Intensive Care Units (NICUs). With a global incidence of approximately 2,824 cases per 100,000 live births and an attributable mortality rate of 17.6%, sepsis poses a significant **burden on healthcare systems worldwide**. Early detection and intervention are critical, as the **"golden hour"** for administering antibiotics is within the first hour after the decision to treat. However, current diagnostic methods often rely on laboratory tests, which can be time-consuming and may delay treatment.

This challenge is being addressed in collaboration with the General University Hospital of Patras, Greece, a leading healthcare institution with a specialised NICU that manages high-risk neonates. The hospital serves as a regional referral centre, providing advanced care for preterm and critically ill newborns.

1.2 General Problem to Be Solved

Neonatal sepsis remains a critical diagnostic and therapeutic challenge in NICUs, particularly at institutions like the General University Hospital of Patras, which manages high-risk neonates, including preterm infants and those with complex medical conditions. The current standard for sepsis detection relies on a **combination of clinical signs** (temperature instability, respiratory distress) and **laboratory tests** (blood cultures, CRP, procalcitonin). However, these methods suffer from significant limitations:

1. **Delayed Diagnosis:** Blood cultures, the gold standard for confirming sepsis, require 24–48 hours for results, delaying life-saving treatment. In contrast, the "golden hour" rule emphasises that antibiotics must be administered within the first hour of suspicion to reduce mortality.
2. **Invasive Procedures:** Repeated blood draws for laboratory tests cause discomfort and increase the risk of anaemia, particularly in fragile preterm neonates.
3. **Non-Specific Symptoms:** Early sepsis often mimics other conditions (e.g., respiratory distress syndrome), making clinical assessment unreliable without confirmatory tests.

For the hospital, these challenges translate into:

1. **Increased workload** for NICU staff, who must monitor neonates closely for subtle signs of deterioration.
2. **Higher healthcare costs** are due to prolonged hospital stays and extended antibiotic use in cases of delayed or uncertain diagnosis.
3. **Preventable morbidity and mortality**, as late treatment escalates the risk of septic shock, organ failure, and long-term neurodevelopmental impairments.

Thus, there is an urgent need for a **real-time, non-invasive, and accurate tool** to detect sepsis before clinical symptoms manifest. Such a solution would empower clinicians at the hospital to intervene earlier, optimise antibiotic use, and ultimately **improve survival and outcomes** for vulnerable newborns.

1.3 Summary of the Proposed Solution

To address this challenge, our solution leverages **artificial intelligence** and **thermal imaging** technology to predict neonatal sepsis early. By continuously monitoring neonates in the NICU using a pocket thermography camera, we capture **thermal and visual images** every minute over a 7-day period. The system analyses these images to detect subtle **changes in peripheral temperature patterns**, which are indicative of sepsis-related vascular dynamics (**centralisation of temperature** with cold peripheries). This non-invasive approach aims to identify sepsis before clinical or laboratory signs appear, allowing for early antibiotic administration and reducing mortality and morbidity rates. The device is not meant to diagnose sepsis by itself, but to raise an alarm when sepsis symptoms appear, **leaving the final decision to the doctor in charge**.

The integration of AI with thermal imaging offers a promising, scalable solution to enhance sepsis prediction in NICUs globally.

2 Proposal of solution

2.1 Objectives

The primary objective of this project is to develop a **non-invasive, real-time monitoring system** capable of detecting early signs of neonatal sepsis using thermal imaging and AI. This tool aims to assist clinicians by continuously analysing peripheral and central body temperature patterns in newborns to identify sepsis-associated anomalies before overt clinical symptoms appear.

Specific goals include:

1. Reducing diagnostic delay by identifying sepsis during the early onset phase.
2. Minimising the need for invasive diagnostic procedures.
3. Providing a clinical decision support system that integrates seamlessly into NICU workflows.

2.2 AI Methodology

The project uses a combination of **deep learning** and **computer vision** techniques to detect early signs of neonatal sepsis from thermal images.

- **Pose Detection** is handled using the **HRNet-W48** deep learning model, pre-trained for human keypoint detection and adapted to identify neonatal body parts (especially joints and face parts) in visible images. These keypoints are used to extract temperature data from thermal images.
- **Modality Alignment** is achieved through **homography transformations**, aligning thermal and visible images using manually selected points. This ensures anatomical accuracy when transferring keypoints between images.
- A **quality control system** filters out images where important keypoints are missing, obstructed, or temperature values fall outside physiological ranges (28–42°C).
- **Temperature analysis** computes the difference between core (Zone 1) and peripheral (Zone 3) body temperatures. Gradients above 3.5°C or temperatures exceeding 38.5°C indicate potential sepsis.
- **Temporal trend analysis** tracks these values over time. Alerts are triggered if abnormal patterns persist for several consecutive readings, enabling early clinical intervention.

2.3 Project planning and methodology

To ensure efficient progress and specialization, the team was divided into **three main working groups**, each focusing on a critical part of the pipeline:

1. **Modality Alignment Group:** Their work ensured that anatomical keypoints from RGB images could be accurately mapped onto thermal frames, forming the foundation for all downstream processing.
2. **Pose Detection Group:** This team focused on choosing, adapting and deploying the posture detection model (in this case, HRNet-W32 architecture) for neonatal keypoint detection.
3. **Rejection Mechanism:** This team focused on implementing quality control checks to reject images that lacked sufficient visibility or produced unreliable temperature readings.
4. **Validation and Results Group:** This team handled the design and execution of experiments using selected patient data (notably cases 36, 47, and 48). Their responsibilities included comparing model outputs with nurse-documented fever events and their human annotations of keypoints, identifying performance gaps, and generating qualitative and quantitative evaluations.

2.4 Ethical and Regulatory Issues

Given the deployment in **high-risk clinical environments**, this project addresses multiple ethical and legal dimensions.

2.4.1 Data Privacy & Consent

- The system handles **sensitive biometric data** (thermal & RGB imagery of neonates).

- Full compliance with **GDPR** and the **European Health Data Space (EHDS)** is enforced.

2.4.2 Regulatory Classification

Under the **EU AI Act**, this system qualifies as a **high-risk AI application** (Annexe III, Section 5(a)), triggering requirements for:

- Risk management and documentation
- Human oversight
- Post-market monitoring and feedback loops

2.4.3 Ethical Risks and Mitigations

- **Automation Bias:** Human-in-the-loop design ensures the final diagnosis rests with clinicians. Alerts are accompanied by confidence scores and visual explanations to support clinician judgment (Explainable AI).
- **Algorithmic Bias:** Ongoing analysis of model fairness across ethnic and clinical subgroups; plans include stratified performance audits and dataset diversification.
- **Moral Responsibility:** A clear **shared accountability model** is defined between engineers (safe design, transparency), clinicians (final decisions), and institutions (oversight and deployment).
- **Surveillance Concerns:** To counter the risk of “normalisation of surveillance”, the system is designed for transparency, with restricted data use strictly for clinical or research purposes only.

3 Implementation of the solution

3.1 System Overview

The developed device is an AI application designed to assist healthcare professionals in the early detection of sepsis in neonatal patients through automated thermal image analysis. The system operates as a clinical decision support tool that processes dual-modality imaging data (thermal and visible spectrum) to identify temperature distribution patterns indicative of sepsis onset in newborns within neonatal intensive care units (NICUs). This device never makes the final decision to diagnose the patient; that is always left to the experienced and knowledgeable doctors.

3.2 Device Architecture and Technical Specifications

3.2.1 Core System Components

The hardware integration layer interfaces with standard thermal imaging cameras commonly deployed in neonatal intensive care environments. The device operates independently of specific hardware manufacturers, accepting thermal image inputs in png formats with typical resolutions of **640x480 pixels** for the thermal images and visible spectrum images, visible images from now on, are processed at **3264x2448 resolution**, creating a **resolution mismatch** that requires specialized alignment techniques to achieve proper spatial correspondence between modalities as we cannot differentiate temperature in visible images, and cannot detect the baby's position in the thermal ones.

The software architecture employs a design built on a machine learning pipeline that processes the image data through **five distinct operational stages**: modality alignment, pose detection, image quality assessment and rejection, temperature analysis, and finally, temporal pattern recognition and alert generation with clinical interface integration.

1. Modality Alignment
2. Pose Detection
3. Image Quality Assessment and Rejection Mechanisms
4. Temperature Analysis
5. Temporal Pattern Recognition and Alert Generation System

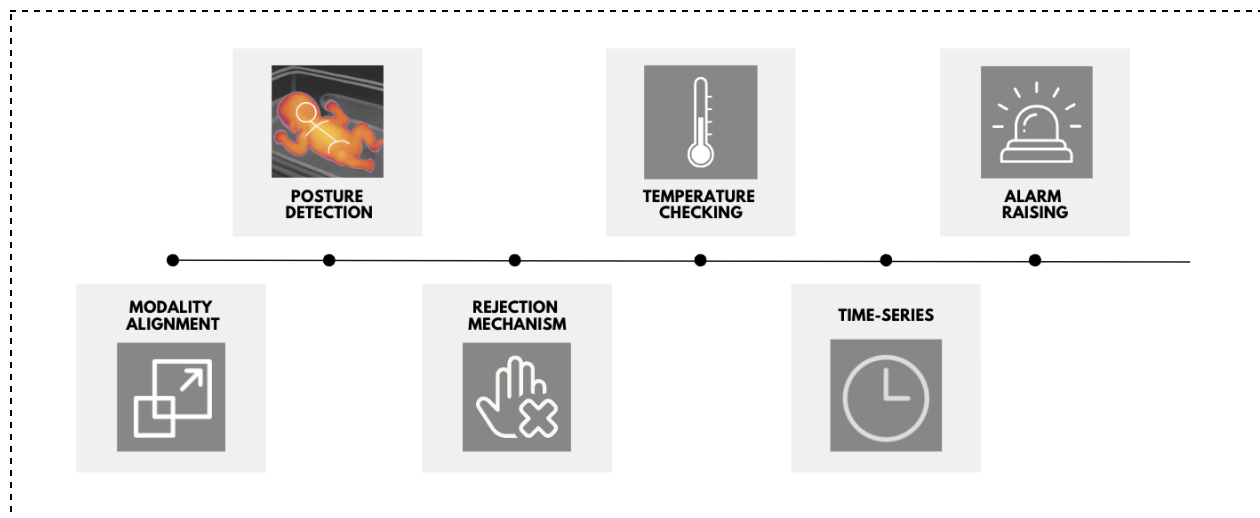


Figure 1: Pipeline overview

3.2.2 Technical Implementation Framework

The system is implemented using **Python-based machine learning frameworks** with primary dependencies on computer vision libraries, including OpenCV and PIL, alongside deep learning

architectures utilising PyTorch optimised for medical imaging applications. The data processing capabilities handle large-scale thermal imaging datasets with current capacity exceeding 252GB of imaging data. **Pre-trained models** were also used for the pose detection section, because they provide reliable results with less effort than training a whole new model on our non-labelled data.

3.3 Technical Methods and Implementation

Stage 1: Modality Alignment

The fundamental technical challenge addressed by this module stems from the **incompatibility between thermal and visible imaging modalities** due to different spatial resolutions, zoom, angles and cropping. Thermal cameras capture temperature distributions but lack the spatial resolution necessary for **accurate anatomical landmark detection**, while visible spectrum cameras provide detailed anatomical information but **cannot measure temperature distributions**.

Modality alignment done in this way stems from the **lack of information about the cameras** that was given, no information about zoom levels, the cropping, or time variations between thermal and visible images. We tried to analyse the HEIC data of images, but no data was found about these discrepancies. It would have been an optimal way of adjusting the two images if we knew this data, but in the status quo, we don't, so we provided the best image matching we could.

The technical solution employs **computer vision-based spatial** correspondence establishment through homography transformation implementation. This process utilizes **point-based correspondence** detection, between two images, the thermal and its corresponding visible image, now we manually select a multitude of points in strategic positions like corners, joints, the silhouette of the baby to match those two images and generate a matrix that permits us to warp all dataset images in the same way so the matching of pose detection points to temperature data can be nearly 1 to 1.

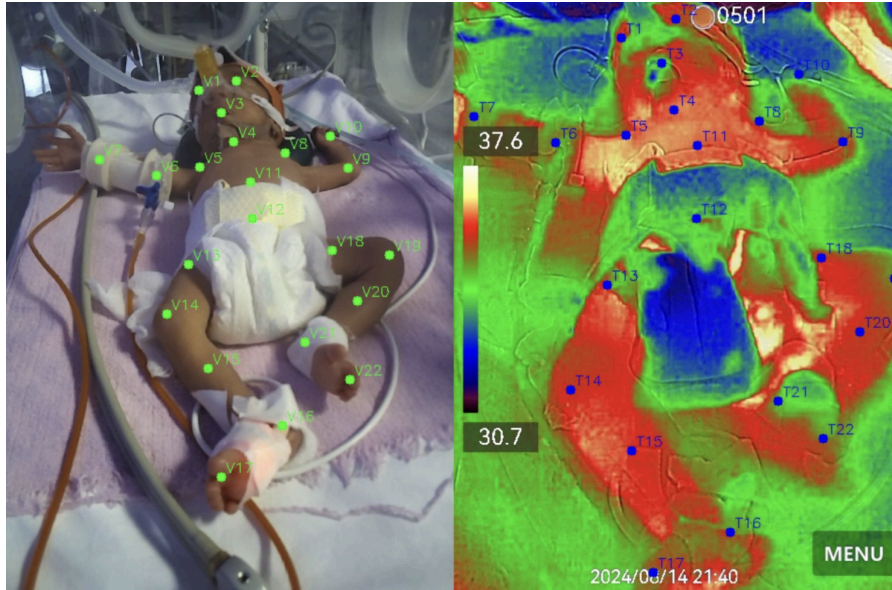


Figure 2: Example of point matching for modality alignment

Image warping techniques implement **perspective transformation** with **bilinear interpolation** for sub-pixel accuracy. Temporal window synchronisation maintains physiological state consistency by aligning image pairs captured within defined temporal windows of 'approximately two seconds'. Quality assessment of alignment utilises **normalised cross-correlation** to verify spatial correspondence accuracy between thermal and visible images. This process was applied two times in the final dataset, first with the first type of thermal cameras, where all images were vertical as the style of figure 2, and finally on the rest of the dataset, shot on a different camera, including both horizontal and vertical images.

This is a crucial part of the pipeline as it is the first step and defines the quality of the updated data; if matching is done incorrectly, it could mean the total failure of the project. In our case, we obtained **good matching results** that were useful and didn't harm the rest of the pipeline.

Stage 2: Pose Detection

Once we have the images optimally cropped (meaning the thermal image and the visible image match almost perfectly), we need to apply the **posture detection** to the visible image to extract useful pixels (**keypoints**) that we will use to check temperature in the thermal image.

The system incorporates **HRNet (High-Resolution Network)** architecture, specifically the **HRNet-W48** variant, adapted for neonatal anatomical detection. This pre-trained model was selected for its ability to detect infant and baby posture points while maintaining **high-resolution feature maps** throughout the network, enabling precise localisation of small anatomical features critical in neonatal subjects where body proportions differ significantly from adult populations. This model was preferred over Mediapipe Pose from Google due to its robustness across different infant positions and camera angles.

The model processes RGB images (visible cropped images from the modality alignment), outputting **17 keypoint coordinates** (17 pixels) with associated confidence scores (scores from 0 to 1). Zone 1 identification focuses on head regions, including nose and ears, along with **chest areas** represented by shoulder keypoints. Zone 2: arms and legs. Zone 3 identification targets **hand and foot keypoints**, which are mapped from wrist and ankle predictions due to the pre-trained model's original training parameters. Confidence thresholding applies a minimum confidence score of **0.7** for reliable keypoint detection. That means we only rely on the output if we are almost sure that the desired keypoint **exists and is correctly located in the body**.

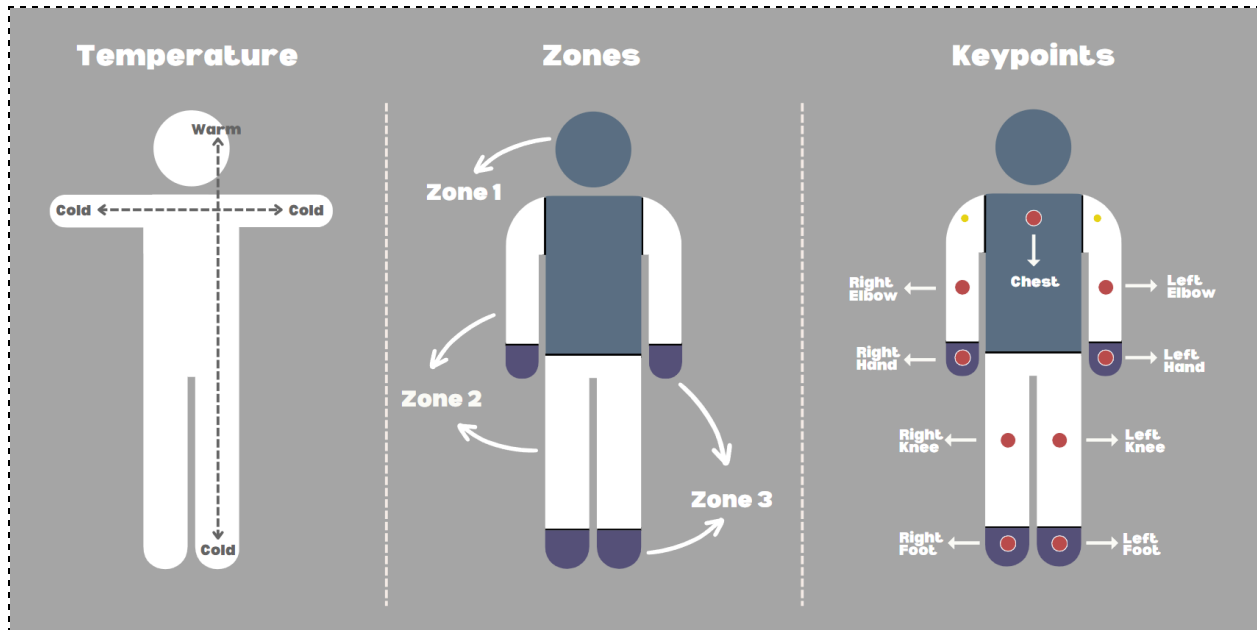


Figure 3: Example of zones and keypoints

A notable technical limitation involves the model's **pre-trained keypoint mapping**: The model is focused on detecting joints in the body (knees, elbows, ankles, shoulders...). Since we want to extract the keypoints shown in Figure 3, we need to make some concessions:

1. **Wrist and ankle detections** must be translated to the **hand and foot regions**. This means for zone 3, we need to use the joints (wrist and ankle) over the actual foot and hands.
2. Since the **chest** (where we want to extract zone 1) is not directly an output of the model, we extract the **middle pixel between both shoulders**. Sometimes shoulders are not detected (because of camera angles or baby posture), and therefore we can't compute the bisector between them. In these cases, we select as zone 1 the **visible shoulder** if it exists (using a **0.7 confidence threshold**).

Stage 3: Image Quality Assessment and Rejection Mechanisms

The quality control framework implements multiple assessment criteria to ensure reliable temperature measurements through **anatomical completeness assessment** and **technical**

quality metrics validation. Anatomical completeness assessment verifies that essential body regions, including head, **chest, and extremities**, remain visible within the imaging field while detecting and handling partial occlusion scenarios. The system assesses subject positioning relative to optimal measurement angles and requires a minimum of **one Zone 1 and one Zone 3 point** to be detected; without these points, a temperature difference could not be measured.

Temperature range validation ensures measurements fall within physiological bounds of **28°C to 42°C**. If higher or lower temperatures are detected, the image, due to its abnormal quality, is discarded — the baby might not be there, or the baby might be too hot, which is probably a wrong temperature reading.

Stage 4: Temperature Analysis

The **temperature analysis module** represents the clinical core of the system, implementing sophisticated algorithms designed to detect **sepsis-indicative thermal patterns** through regional temperature extraction and clinical pattern recognition. **Zone 1 core temperature calculation** employs Gaussian-weighted averaging across head and chest regions using a 15×15 pixel kernel with a sigma value of 5. The system applies outlier removal and implements **preference weighting**, favouring chest measurements at 0.7 weighting compared to shoulder measurements at 0.3 weighting. Same done with Zone 3 key points.

Temperature gradient analysis calculates core-to-peripheral temperature differentials as the primary diagnostic indicator. The system identifies two primary sepsis-associated thermal presentations: generalised hyperthermia detection when Zone 1 temperatures exceed 38.5°C, and central-peripheral temperature dissociation when the absolute difference between Zone 1 and Zone 3 exceeds 3.0-3.5°C with temporal persistence requirements exceeding a threshold of consecutive measurements, manually selected.

Stage 5: Temporal Pattern Recognition & Alert Generation System

A **time series map** of key temperature points is generated for each patient, focusing on comparisons between **Zone 1 (forehead)** and **Zone 3 (foot)**. **Baseline temperatures** are established using median values from initial measurement windows, forming the base for standard deviation analysis.

The system identifies potential **fever conditions** based on a combination of criteria: a sustained temperature difference between Zone 1 and Zone 3 exceeding 3.5°C for more than 5 consecutive measurements – all thresholds manually selected – or persistence of abnormal patterns through a time period. When these thresholds are met, alerts are triggered, currently only on our code output. We will discuss some possible improvements for this stage in the proposed technical improvements tab.

Validation Framework

For our validation, we focused on babies 36, 47, and 48. The results will be discussed in the results section.

3.4 Regulatory Compliance and Safety Implementation

3.4.1 Safety Mechanisms

Ideally, we would keep developing the model safely and legally with the following ideas in mind:

- **Fail-safe operation design** would incorporate a ‘watchdog’ timer implementation for continuous system health monitoring with automatic fallback to alert states during system failures.
- The **audit trail system** would maintain comprehensive logging using structured JSON formatting with immutable log storage protected by cryptographic hashing. Real-time log analysis enables anomaly detection.
- **Data security implementation** would utilise **AES-256 encryption** for data at rest protection and **TLS protocols** for data transmission security. **Role-based access control** systems would implement **JWT token authentication** with automated security scanning and vulnerability assessment protocols.

3.5 Technical Limitations and Future Enhancements

3.5.1 Current Constraints

Hardware dependencies include thermal camera calibration requirements that vary across different manufacturer specifications and **environmental sensitivity characteristics** of thermal sensors that require controlled ambient conditions. **Processing power requirements** for real-time operation necessitate dedicated computational resources for optimal performance in clinical environments.

Algorithm limitations encompass **pose detection accuracy** challenges in extreme infant positioning scenarios and **motion artefact handling** complications in thermal imaging applications.

3.5.2 Proposed Technical Improvements

Enhanced algorithm development, like **deep learning-based image alignment** techniques replacing traditional homography methods and **attention mechanism implementation** for improved keypoint detection accuracy, or better control of our hardware so that matching never becomes a problem. Additionally, a possible fine-tuning of **HRNet** to map hands and feet

(instead of wrists and ankles) and central Zone 1 points, such as the chest (instead of the bisector of the two shoulders).

For the time series, **advanced time series** using **LSTM networks** could enhance the model's prediction capabilities. Alerts should be distributed via a **Redis-based message queue** for efficient delivery across multiple clinical endpoints. Simultaneously, a **WebSocket** would push real-time updates to monitoring dashboards for continuous clinical oversight. Each alert should be classified by a **risk-scoring algorithm** based on the temperature gradient magnitude, temporal persistence, and rate of change. Risk levels could be defined in the following manner:

- Low Risk: Gradients $> 2.0^{\circ}\text{C}$
- Moderate Risk: Gradients $> 3.5^{\circ}\text{C}$ with persistence > 15 minutes
- High Risk: Composite scores indicating urgent clinical intervention

This technical implementation represents a comprehensive approach to neonatal sepsis detection using thermal images.

4 Results

Patient Selection: Analysed 71 neonates; focused on 3 sepsis+fever cases (IDs 36, 47, 48) with documented fever onset timing.

4.1 Methodology

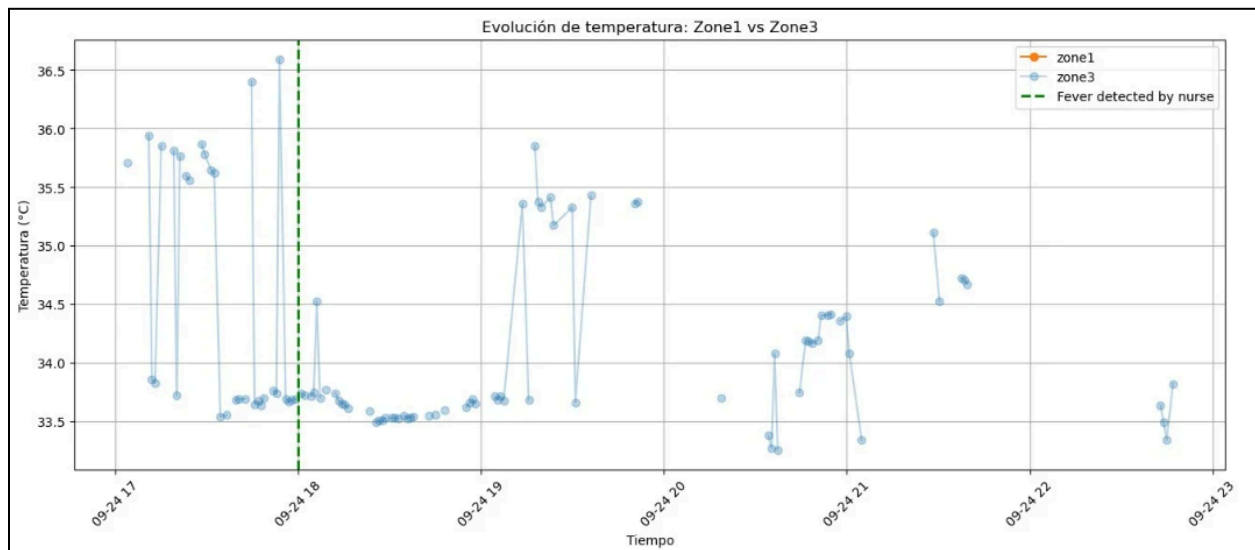
1. **Core Mechanism:** The Temperature monitoring system triggers alarms when inter-zone temperature differences exceed predefined thresholds.
2. **Control Benchmark:** Human annotations simulating ideal measurement points in each thermal image (pre/post nurse-detected fever timing).
3. **Model Threshold:** Alarm requires ≥ 2 consecutive abnormal readings to prevent false alerts from single-point anomalies.
4. **Validation:** Model outputs compared against:
 - Nurse-documented fever events
 - Retrospective human annotations

4.2 Results of the Experiments

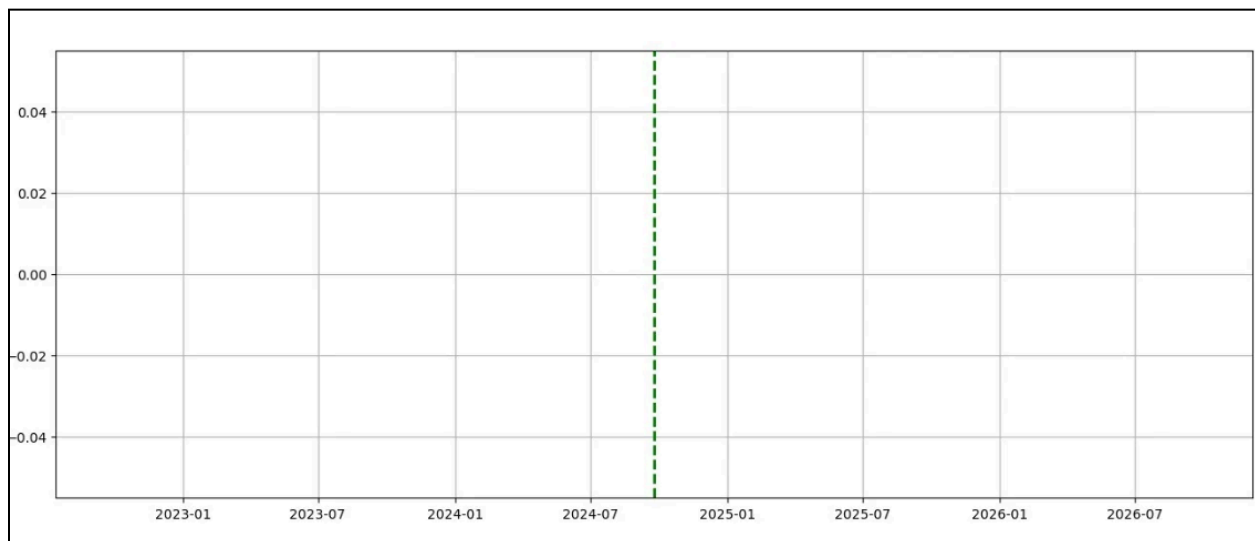
4.2.1 Case 36: Posture Detection Failure

Observation:

- Critical Gap: The Model failed to detect measurable points during the nurse-identified fever period
- Model Decision: No alarm triggered (insufficient consecutive readings)



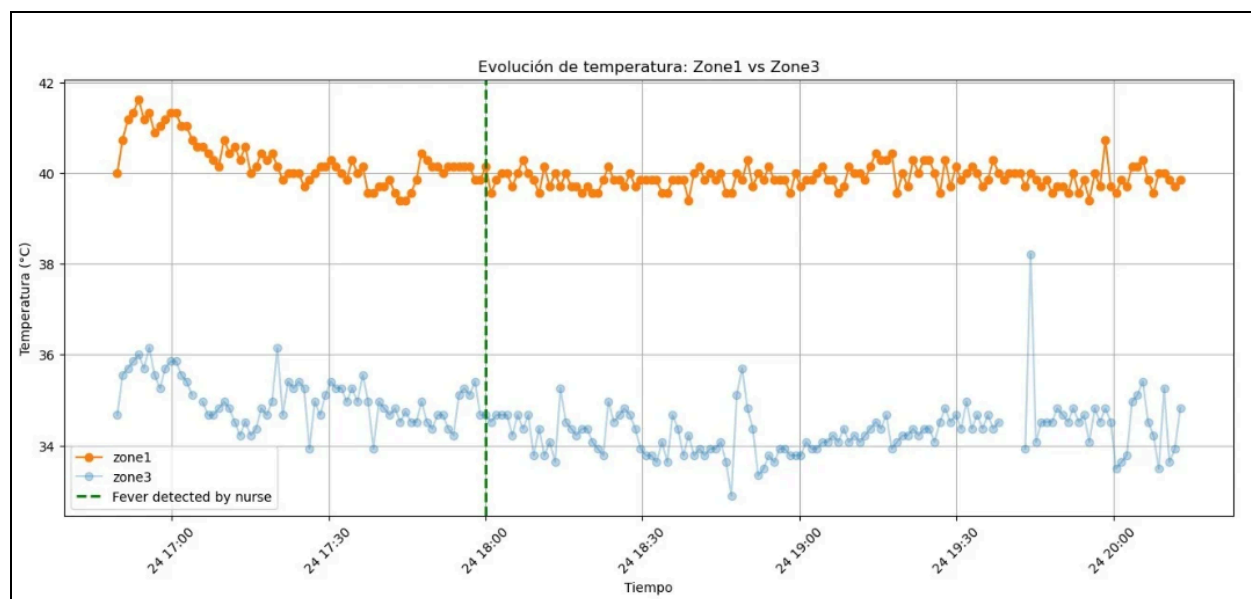
Model Result



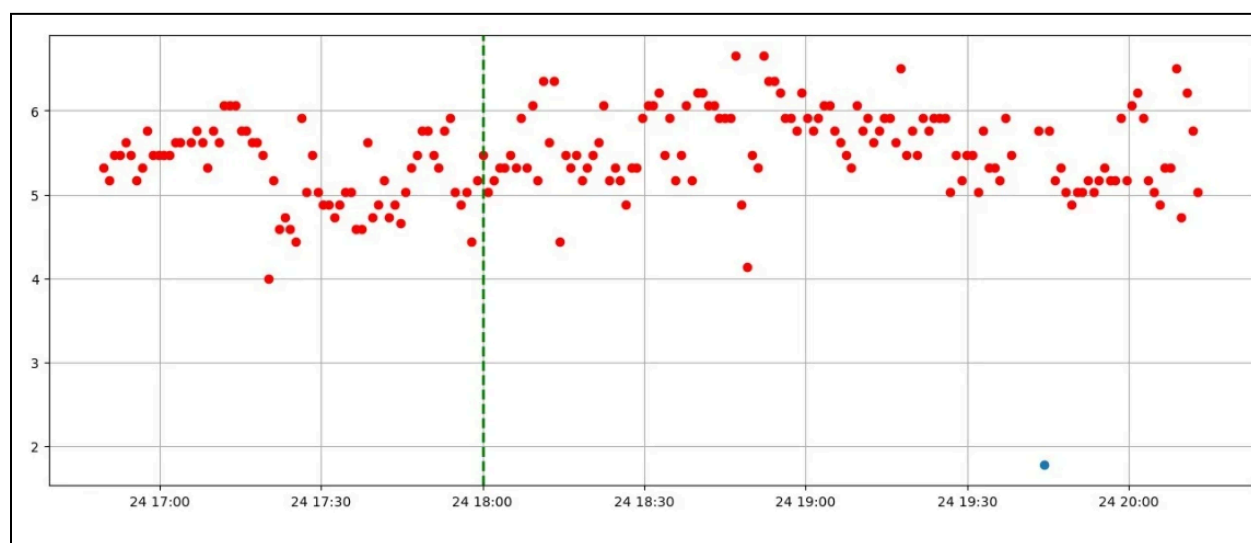
Model Result Differences of temperature

Human Annotation Analysis:

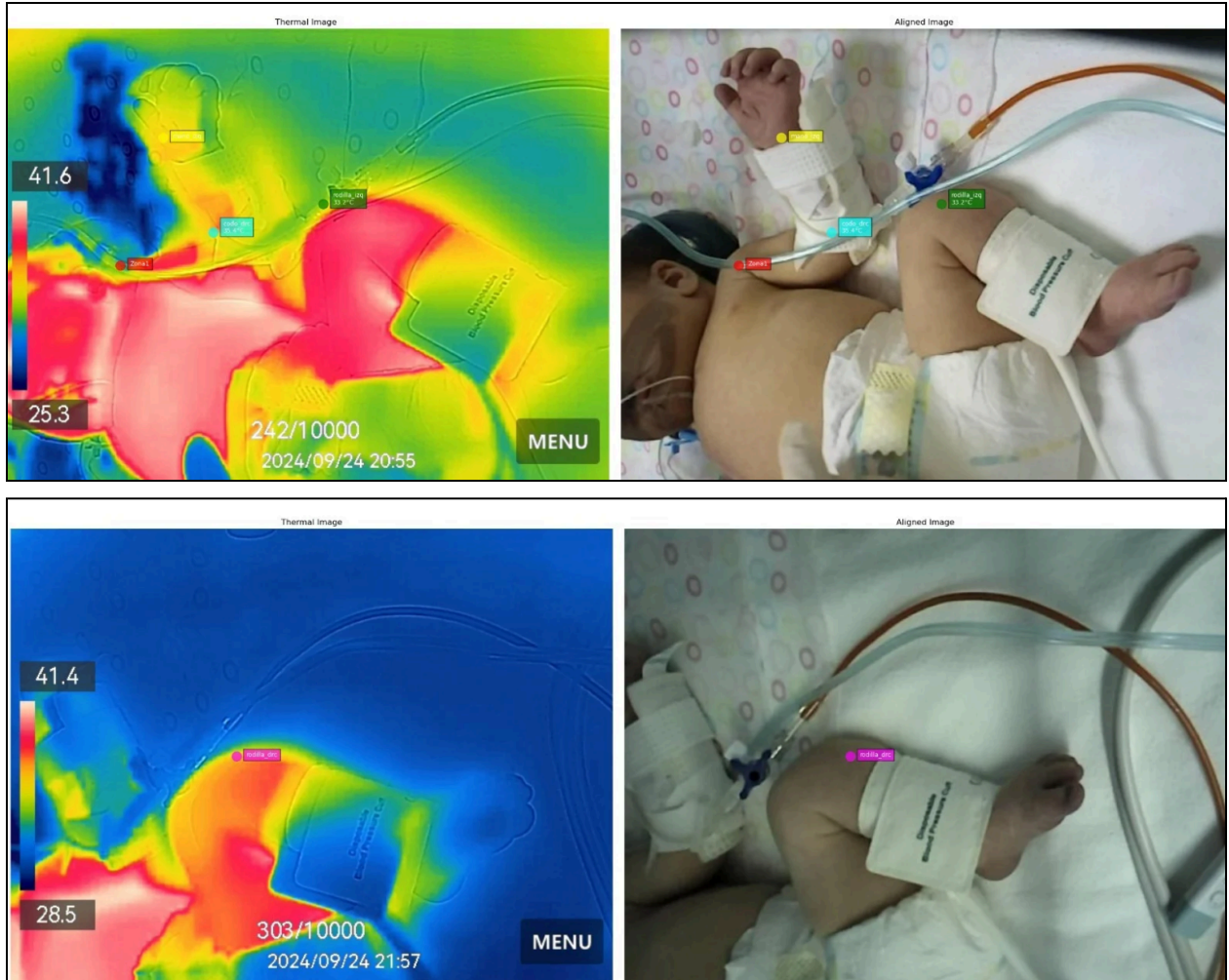
- Revealed fever could have been identified earlier
- Failure Cause: The Posture detection system is unable to capture valid measurement points
- Implication: Technical limitation prevented potential early detection



Human Annotation



Difference of temperatures Human Annotation



Objects can be observed that interfere with temperature measurement, as seen in the first image, where the presence of the tube causes that point to be excluded, resulting in the loss of many data points. On the other hand, there are images where only zone 2 or 3 is present, but not zone 1, as in the last image.

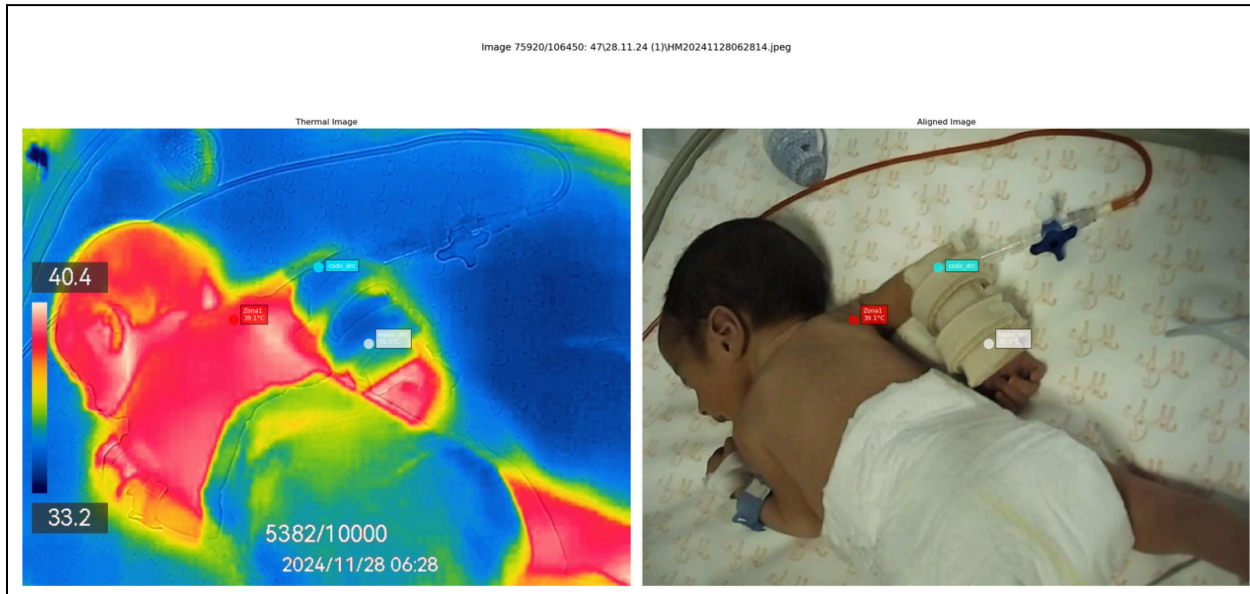
4.2.2 Case 47: Contextual False Alarm

Observation:

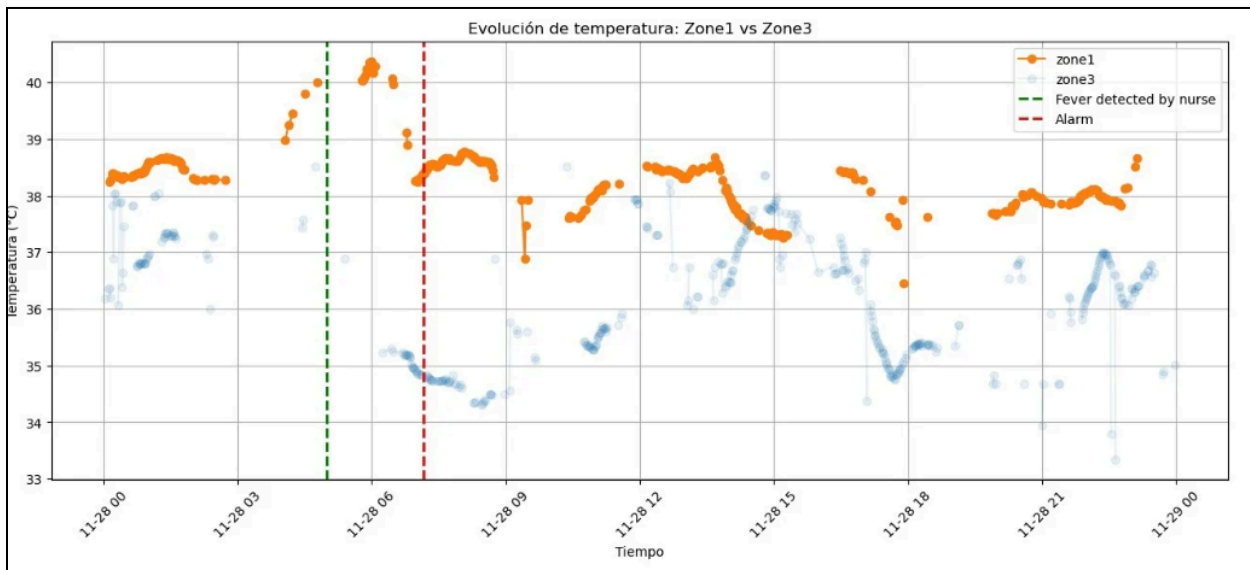
- The alarm triggered a bit after the nurse detected the fever..
- The graph showed Zone 1 $\uparrow$ / Zone 3 $\downarrow$ (expected sepsis pattern). But not all that glitters is gold.

In this specific case, we have bad news. Because looking at the human annotation, we realise that no matter how well the model performs, even achieving human-level performance, we are unable to detect the fever before the nurse.

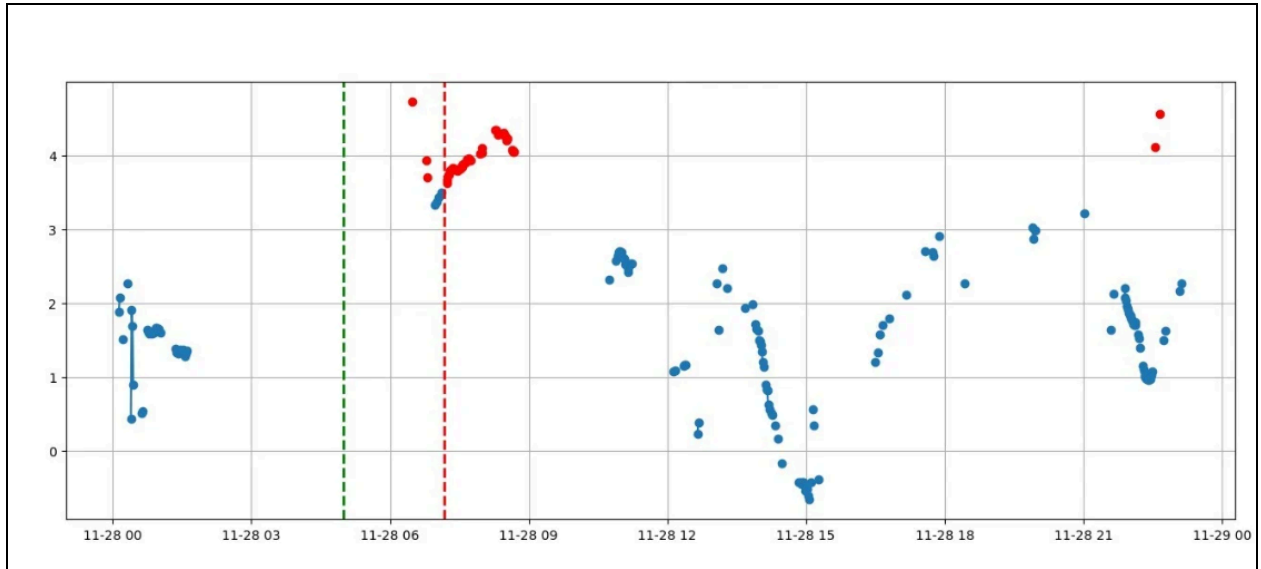
This difference between the performance of the model and the performance of humans is strange, so let's inspect what the model is doing in those cases. The images show how the colour of the bandage is very similar to the colour of the baby, which leads to errors in the rejection mechanism. This way we're measuring the temperature on the bandage, which is considerably lower than the ideal measurement. This explains the pattern we have seen where the temperature of zone 1 increases and the zone 3 decreases at the same time.



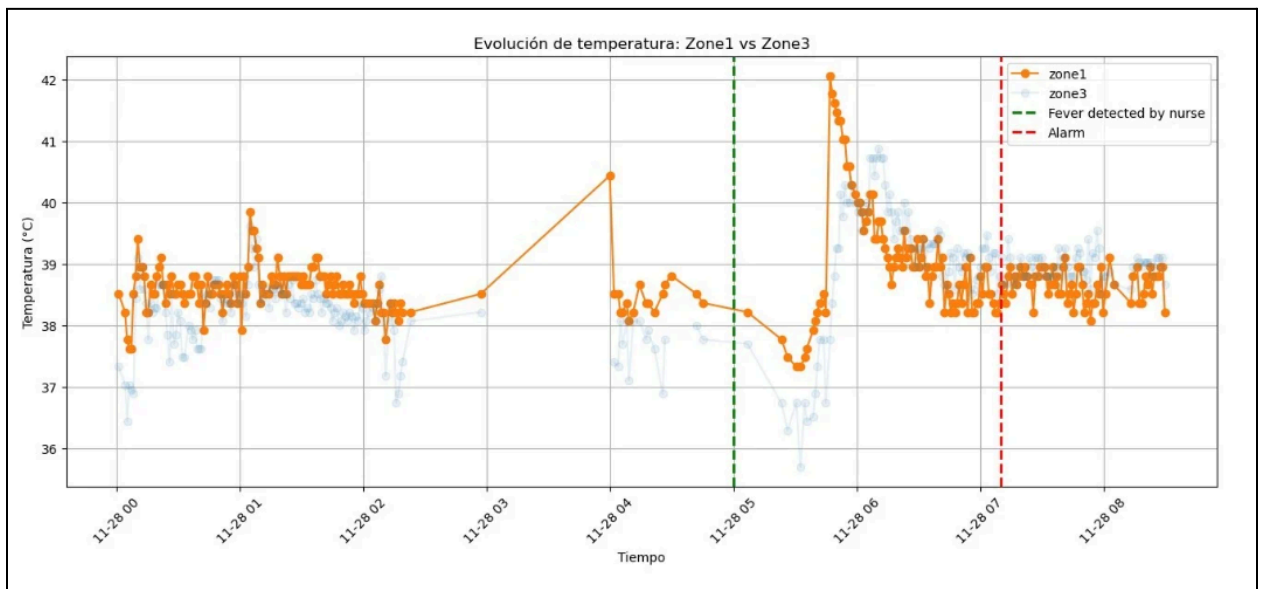
Images of the model



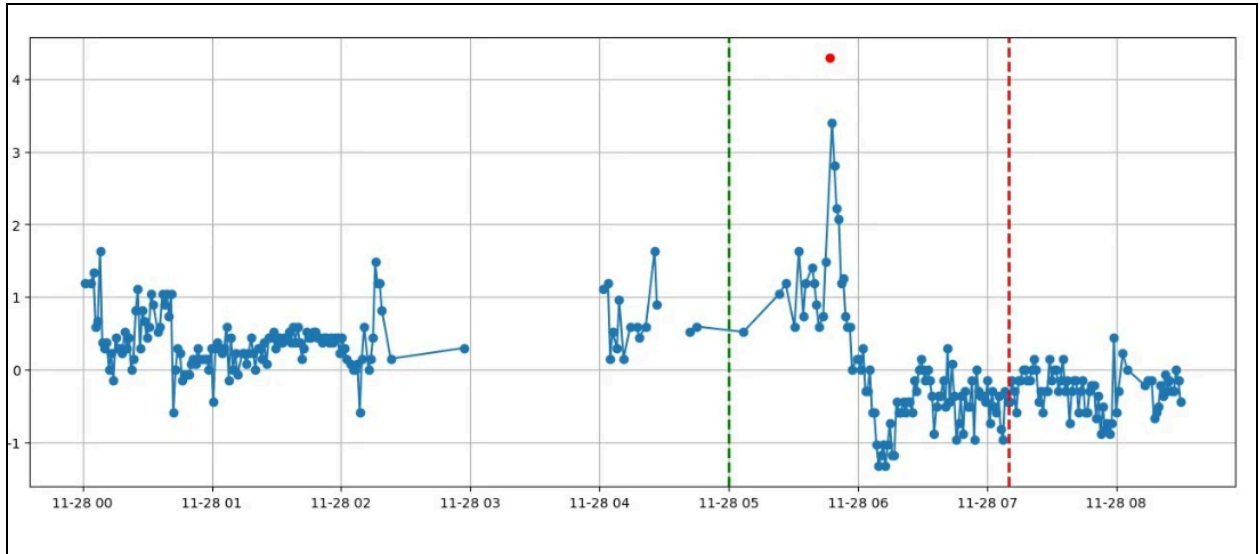
Model Result



Model Differences



Human Annotations

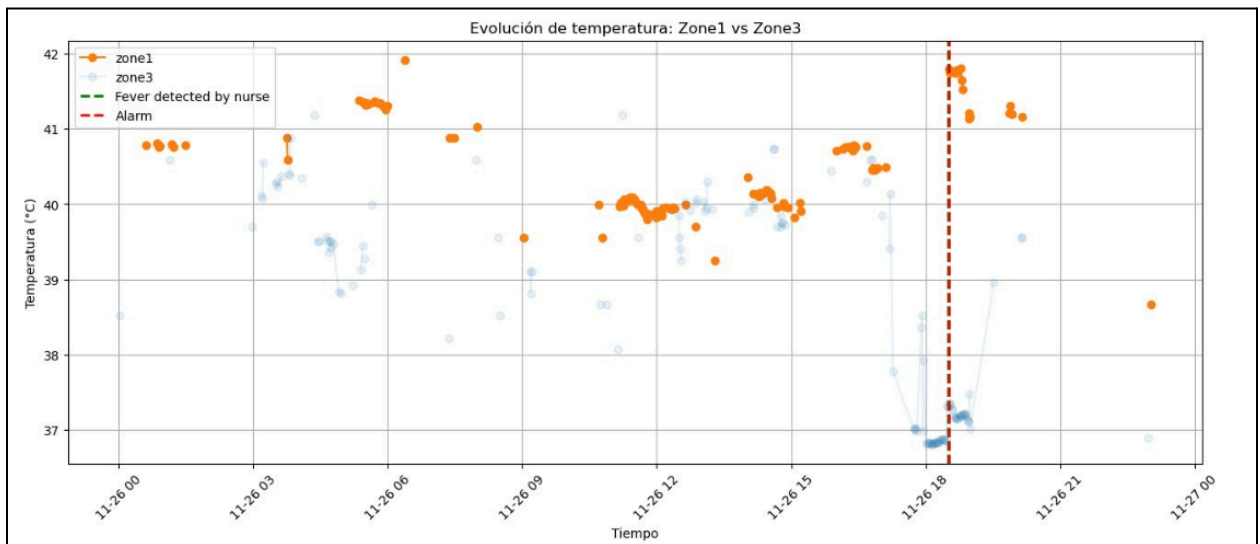


Human Differences

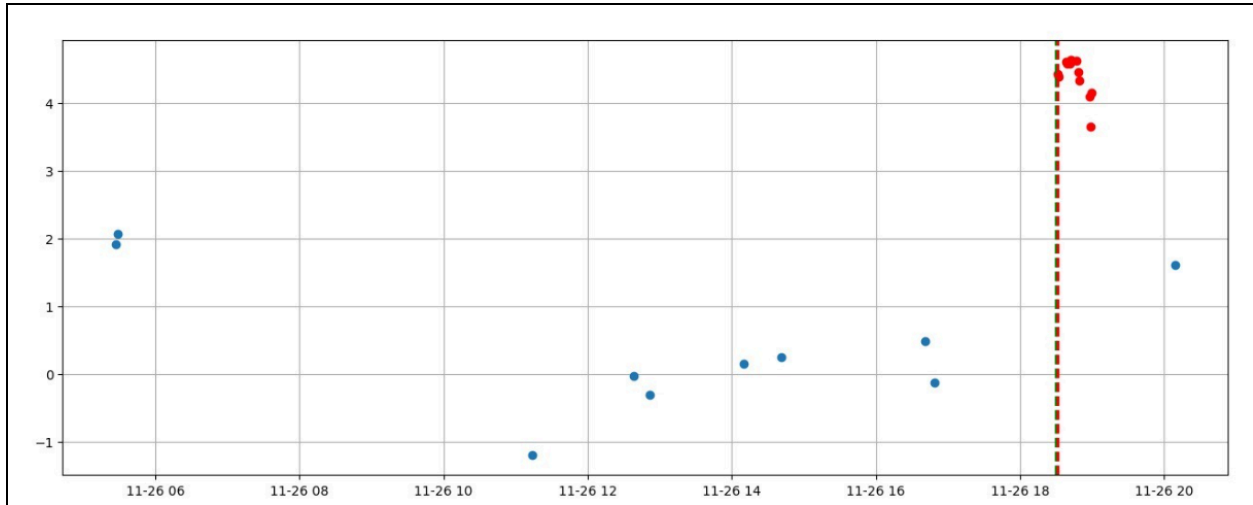
4.2.3 Case 48: Optimal Detection

Observation:

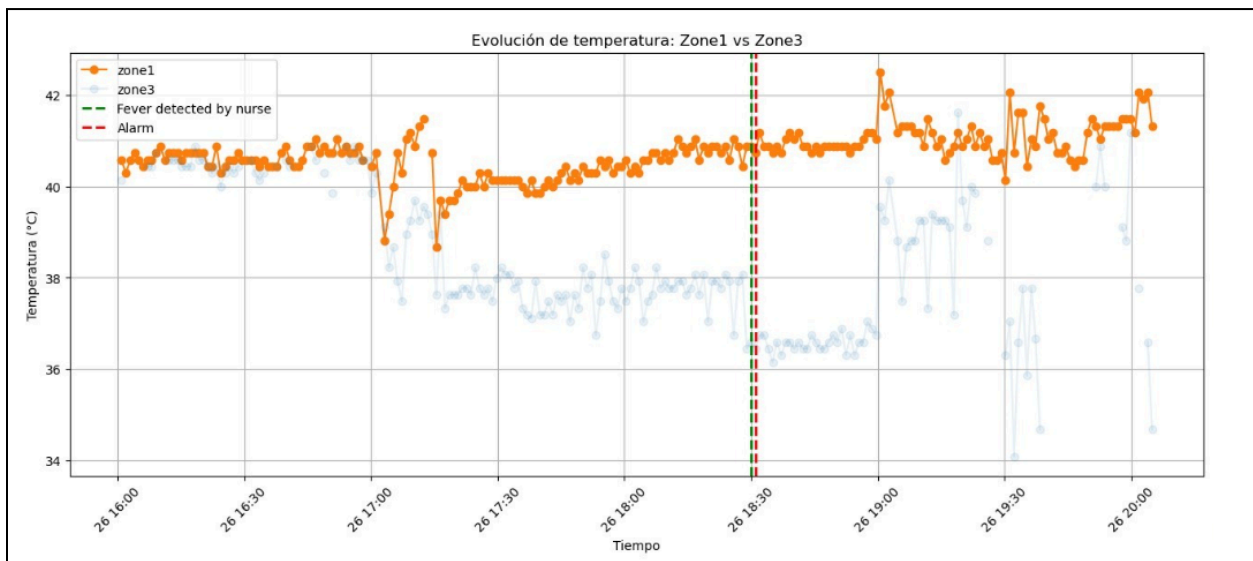
- Model-detected fever time = Nurse-recorded time (18:30).
- Temperature trends aligned perfectly with human annotations.
- Significance: Demonstrated model reliability under ideal conditions (no artifacts).



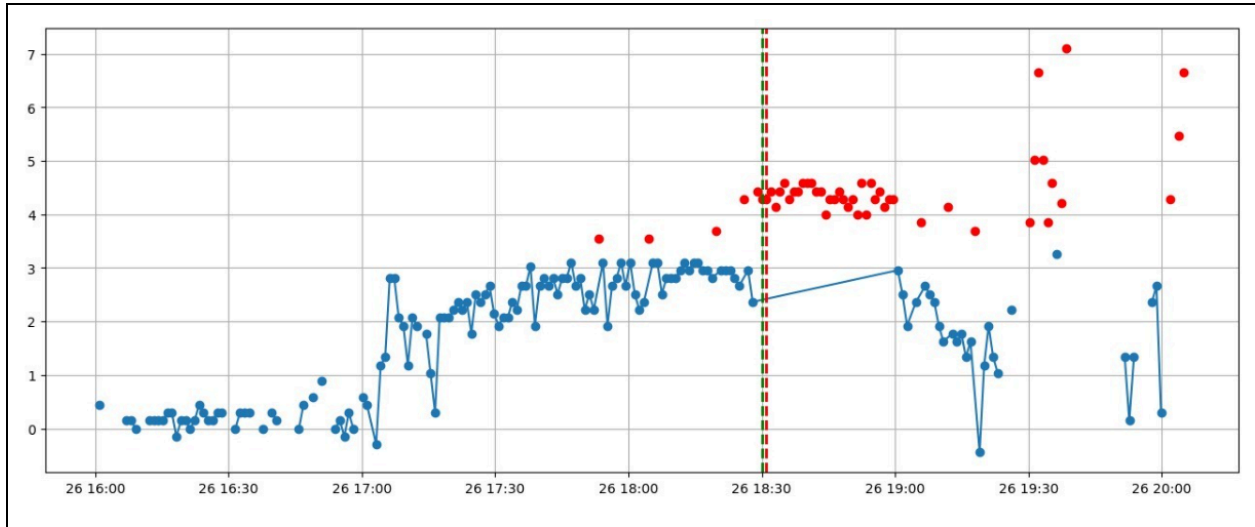
Model Result



Model Differences



Human Annotation



Human Annotations Differences

Metric	Case 36	Case 47	Case 48
Detection Outcome	Missed opportunity	Incorrect* (with artifact)	Correct
Human-Model Gap	High	Moderate	None
Key Limitation	Posture detection	Color interference	None

4.3 Qualitative & Quantitative analysis of the results

4.3.1 Qualitative Findings

Critical Success Factors:

- **Image Quality Dependencies:** The model's reliability is heavily contingent on thermal image clarity and colour contrast. Bandages, sensors, and other medical artefacts create significant interference that can compromise temperature measurement accuracy.
- **Posture Detection Challenges:** The system's ability to identify valid measurement zones is critically dependent on patient positioning and the absence of obstructing objects (tubes, bandages, sensors).

Clinical Performance Patterns:

- **Case 36 Analysis:** Complete detection failure due to the posture detection system's inability to capture sufficient valid measurement points during the critical fever period. Human annotations revealed that earlier detection was theoretically possible, highlighting a significant technical gap.

- **Case 47 Analysis:** The model exhibited the expected sepsis pattern (Zone 1↑/Zone 3↓) but was compromised by bandage colour similarity to skin tone, leading to temperature measurements on the bandage rather than skin. This artifact explains the observed temperature differential pattern but renders the detection clinically unreliable.
- **Case 48 Analysis:** Demonstrated optimal performance with perfect temporal alignment between model detection and nurse documentation, validating the system's capability under ideal conditions.

4.3.2 Key Findings

The thermal monitoring system achieved **optimal performance under ideal conditions**, as evidenced by Case 48, where model detection perfectly aligned with nurse-documented fever onset. This validates the fundamental approach of using **inter-zone temperature differentials** to identify febrile episodes. However, our analysis revealed **critical dependencies on image quality and measurement artifact management** that significantly impact clinical reliability.

The most significant limitation identified was the system's vulnerability to measurement interference from common neonatal care equipment. Case 36 demonstrated how posture detection failures can result in complete missed detection opportunities, while Case 47 revealed how bandage color interference can create false patterns that mimic sepsis indicators but provide clinically unreliable information.

5 Conclusions and open challenges

The project has developed a novel system for detecting sepsis/fever in newborns by integrating thermal and visible imaging.

The system showed promise by accurately detecting fever in one case (Case 48) at the same time as clinical staff. However, it also encountered significant challenges. In Case 36, the system failed to detect the fever due to posture detection issues, and in Case 47, it produced a false alarm caused by artefacts such as bandages.

While the system shows potential for detecting sepsis and fever in newborns, its performance varies across cases. **Addressing posture detection issues** and **reducing false alarms** will be critical for enhancing its reliability and applicability in clinical settings.

5.1 Key Achievements

In the **Modality Alignment** stage, we successfully resolved the mismatches between thermal and RGB images—differences in resolution, zoom, angle and cropping—without relying on any camera metadata. By implementing a **manual, point-based** correspondence protocol grounded in homography transformations, we achieved precise spatial alignment of landmarks, creating a reliable foundation for all subsequent processing steps.

During **Pose Detection**, we adapted the **HRNet-W32**—which has been primarily trained with pictures of adults—to identify **seventeen anatomical keypoints** on neonates despite their unique proportions. Operating at a resolution of 256×192 pixels with a confidence threshold of 0.70, the system consistently localised the head, chest, limbs and extremities, while automatically dismissing obstructed or low-confidence detections.

Also in the **Temperature Analysis** stage, we've managed to extract regional temperatures using **Gaussian-weighted averaging** and calculate **core-to-peripheral gradients** to identify sepsis indicators, such as hyperthermia ($>38.5^{\circ}\text{C}$ in Zone 1) or temperature dissociation (Zone 1–Zone 3 difference of $3.0\text{--}3.5^{\circ}\text{C}$). This successfully translates thermal data into information doctors can use.

Finally, in the **Temporal Pattern Recognition & Alert Generation** stage, we successfully track temperature trends over time, establishing baselines and triggering alerts based on anomalies (e.g., $>3.5^{\circ}\text{C}$ difference for five consecutive measurements). This capability enhances the system's potential for early intervention.

5.2 Open Challenges

While the system shows strong potential, several obstacles remain to ensure its reliability and clinical adoption.

5.2.1 Dataset

- **Angle Consistency:** All pictures should be taken at a **90° angle** with respect to the baby, with the baby centred in the frame. This standardizes the perspective, ensuring reliable image analysis and comparison across datasets.
- **Camera Calibration:** Cameras must be calibrated before use to ensure that both thermal and RGB images have the **same dimensions**. This alignment is vital for accurate overlay and integration of data from both imaging types.
- **Lighting Uniformity:** All pictures should have consistent lighting conditions. Uniform lighting makes analysis more precise and easier.
- **Movement Between Pictures:** Movement of the baby between thermal and RGB captures can cause misalignment, complicating feature matching across the images. Techniques to minimize or correct for this movement are essential.
- **Not Enough Data on Babies with Sepsis:** The lack of sufficient data on babies with sepsis obstructs the development and evaluation of the model. **Expanding data collection** is necessary to overcome this limitation.

5.2.2 Modality Alignment

- **Automation:** The manual point selection process is labour-intensive; if we wanted the best correspondence matrix for each baby, we would need to go picture by picture selecting points; automating correspondence detection would improve efficiency.

5.2.3 Pose Detection

- **Neonatal Adaptation:** Fine-tuning HRNet-W32 on neonatal-specific data could improve keypoint accuracy.
- **Zone Mapping:** Refining the translation of wrist/ankle keypoints to hand/foot regions could enhance temperature accuracy.

5.2.4 Temperature Analysis

- **Extraction Precision:** Improving temperature extraction in areas with fluctuating brightness or under blue light.

5.2.5 Temporal Pattern Recognition & Alert Generation

- **Clinical Validation:** More trials are required to confirm alert reliability, minimising false positives while ensuring sensitivity to early sepsis signs.

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